

Assignment 3 - Essentials of Project Management & Project Management Planning and Scope

A practical written task to define project scope, plan execution, manage quality, and prepare professional delivery.

Session	Session 3: Essentials of Project Management & Project Management Planning and Scope	Assignment Type	Individual Written Task
Assignment Title	Freelance Project Plan and Delivery Roadmap	Total Marks	20
Submission Format	Written document only - no audio or video	Suggested Length	1-2 pages
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Assignment Brief

One Practical Assignment

Choose a small freelance project related to your service. Create one written Project Plan and Delivery Roadmap that you could share with a client before starting the work.

- Summarize the client requirement, project objective, measurable deliverables, assumptions, and what is in scope and out of scope.
- Break the project into tasks, milestones, deadlines, dependencies, and one key risk with a practical mitigation action.
- Add one client progress-update message plus a short quality, final-delivery, revisions, and project-closure checklist.

Required Written Submission

Final Output	- Mini project plan: requirements, objective, scope, deliverables, tasks, milestones, and timeline. - One project risk with its mitigation action and one professional progress-update message. - Quality and delivery checklist covering review, testing, handoff, revisions, feedback, and closure.
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Evaluation Rubric - 20 Marks

Criteria	Marks	What the trainer will check
Requirements, scope, and deliverables	5	Project requirements are clear; scope boundaries and measurable deliverables prevent confusion and scope creep.
Tasks, timeline, and risk	5	Plan includes logical tasks, milestones, deadlines, dependencies, and a realistic risk response.
Communication, quality, and closure	5	Update message and checklist support professional execution, quality control, delivery, revisions, feedback, and closure.
Formatting and language	5	Submission is organized, concise, readable, and professionally written.

Submission note: The assignment must be completed in writing only. Do not submit voice notes, videos, or presentation recordings.

Executive Summary & Client Requirements

The client currently relies on manual visual spot-checks or legacy inspection methods for quality assurance and operational monitoring. This manual process introduces potential bottlenecks, human oversight errors, and inconsistent defect classification.

The objective of this freelance project is to design, train, and deploy a robust **Computer Vision Inspection System** integrated with the client's existing video or image input feed. The system will automatically classify target objects or defects in real-time, flag anomalies before downstream processing, and log structured inspection data into an accessible dashboard or database.

- ❖ **Input Stream:** Real-time video feed or high-resolution image batches from industrial cameras / capture devices positioned at the inspection station.
- ❖ **Detection Target:** Target objects, components, or surfaces passing through the operational area under specific operational speeds.
- ❖ **Classification Criteria:** Pass (Standard / Acceptable), Defective (Specific structural or surface anomalies), and Obstructed / Unclear.
- ❖ **Processing Constraints:** Real-time edge or cloud inference latency within target threshold; high accuracy under variable ambient lighting conditions.

Scope, Deliverables, and Assumptions

Scope Boundaries

A clear definition of project boundaries prevents scope creep and ensures smooth execution.

In-Scope (Included in Project)	Out-of-Scope (Excluded / Phase 2)
• Custom object detection/classification model trained on client dataset. • Optimized inference script configured for target edge/cloud hardware. • API endpoint or trigger mechanism for system alerts. • Lightweight web dashboard for logging inspection snapshots and metadata. • Comprehensive validation report and deployment documentation.	• Replacement of physical cameras or lighting hardware. • Deep integration with enterprise ERP/WMS systems (scheduled for Phase 2). • Multi-site cloud synchronization infrastructure. • Specialized 3D LiDAR or laser scanning hardware integration. • Direct automated mechanical sorting actuator control.

Measurable Deliverables

- ❖ **Deliverable 1: Trained Model Weights:** Optimized deep learning model weights meeting agreed-upon evaluation metrics (e.g., mAP or accuracy thresholds) on validation data.
- ❖ **Deliverable 2: Deployment Package:** Containerized or modular inference engine configured for the designated target hardware environment.
- ❖ **Deliverable 3: Inspection Dashboard:** Functional web interface or logging utility displaying real-time pass/fail metrics and anomaly logs.
- ❖ **Deliverable 4: Handover & Training Manual:** Documentation detailing model retraining procedures, system architecture, and operational instructions.

Key Project Assumptions

- ❖ The client will provide representative raw or labeled training images/video samples within 3 business days of project kickoff.
- ❖ Stable network connectivity, power, and physical access to camera feeds/testing environments will be maintained throughout evaluation.
- ❖ The client technical stakeholder will be available for scheduled sync meetings and prompt feedback on milestone deliverables.

Project Breakdown, Milestones, and Timeline

The project is structured into 4 sequential phases spanning 20 business days (4 weeks). Each phase concludes with a formal milestone review.

Phase / Task	Wk / Days	Dependencies	Key Deliverable / Milestone	Deadline
Phase 1: Data Audit & Setup • Input feed analysis • Dataset annotation & cleaning	Days 1-5 (Week 1)	None	Milestone 1: Approved Dataset & Pipeline Architecture	Day 5
Phase 2: Model Training • Architecture tuning & training • Hyperparameter optimization	Days 6-10 (Week 2)	Phase 1 Sign-off	Milestone 2: Baseline Model Validation	Day 10
Phase 3: Hardware Integration • Model export & optimization • Stream & API hook setup	Days 11-15 (Week 3)	Phase 2 Sign-off	Milestone 3: Working Prototype on Test Rig	Day 15
Phase 4: Testing & Handover • Live stress testing • Dashboard UI & Documentation	Days 16-20 (Week 4)	Phase 3 Sign-off	Milestone 4: Final System Deployment & Sign-off	Day 20
Project Closure • Knowledge transfer • Final sign-off & feedback	Day 20 (Week 4)	Milestone 4	Project Closure & Handoff	Day 20

Risk Management & Mitigation

CRITICAL RISK IDENTIFIED: Risk: Variable ambient lighting conditions (glare, shifting natural light, or shadows across operating hours) causing false anomaly positives or missed defects.

Impact & Probability: High Impact, Medium Probability.

Practical Mitigation Action: Integrate robust data augmentation strategies during training (incorporating random brightness, contrast, and shadow simulations). Additionally, configure automated preprocessing filters (histogram equalization / adaptive contrast normalization) in the video ingestion pipeline and establish a dedicated Region of Interest (ROI) mask.

Professional Client Progress-Update Message

Below is a sample weekly progress update message ready to be customized and sent via communication channels to the client stakeholder.

Subject: Weekly Status Update: Computer Vision Inspection Project (Week 2)

Hi [Client Name],

I hope you're having a great week!

I'm writing to share our progress report for Week 2 of the Computer Vision project. We have successfully completed Phase 2 on schedule:

- Completed Tasks:
 1. Finished data augmentation and annotation cleaning on the training dataset.
 2. Trained the baseline model, achieving target validation accuracy on the holdout set.
 3. Implemented preprocessing color correction to handle operational lighting variations.
- Planned Focus for Next Week (Week 3):
 1. Export model weights to optimized format for target deployment hardware.
 2. Connect live video stream for latency and throughput benchmarking.
- Action Items / Decisions Needed:
 - Could you confirm our access window for live testing next Tuesday morning?

Let me know if you would like to schedule a quick sync to review preliminary detection logs. Have a wonderful rest of your week!

Best regards,

[Your Name]

Computer Vision Engineer

Quality, Delivery, Revisions, and Closure Checklist

This comprehensive checklist governs quality assurance, client review, testing protocols, revision cycles, and formal project closure.

Category	Checklist Item Description	Status	Responsible
Review & QA	Verify model evaluation metrics exceed agreed threshold against test set.	☐ Pending	Consultant
Review & QA	Perform code linting, security review, and dependency check.	☐ Pending	Consultant
Testing	Conduct live stress test on environment with real input stream.	☐ Pending	Joint (Both)
Final Handoff	Deliver deployment container, model weights, and source code via secure repository.	☐ Pending	Consultant
Revisions	Incorporate up to agreed rounds of client feedback on dashboard UI / thresholds.	☐ Pending	Joint (Both)
Project Closure	Conduct technical knowledge transfer session and sign off handoff document.	☐ Pending	Joint (Both)